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## Detailed Course

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### Introduction and Motivation

#### General Context

**PCA (Principal Component Analysis)** is a **linear latent decomposition** method that identifies the **underlying latent factors** structuring data.

## Main Problem:

Base representations of data are not always optimal. PCA seeks to find a new representation that:

- Reveals the **latent factors** (hidden) governing the process
- Uses **statistical correlation** to identify these factors
- Enables **simplification** of data through justified decimation

## Fundamental Characteristics

PCA exhibits **linear latent factors** and allows several complementary interpretations that we will study.

**Key Point:** PCA belongs to the family of **linear latent models**

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## Formalization

### Maximizing Variance

### Geometric Intuition

**Fundamental Question:** What information should we preserve when projecting?

### Illustrative Example:

Suppose data  $X \subset \Omega$  is almost contained within a 2D plane in a 3D space. The relevant choice is to select a basis  $\{u_1, u_2\}$  for this plane.

**Characterization of the Optimal Basis  $\{u_1, u_2, u_3\}$ :**

- Data **varies most** along directions  $u_1$  and  $u_2$
- Data **varies least** orthogonally to these directions (along  $u_3$ )

## Mathematical Formalization

### Definitions:

Let  $X$  be the dataset,  $\{u_i\}_{i \in \llbracket D \rrbracket}$  a new **orthonormal basis** of  $\mathbb{R}^D$ .

The **first principal component**  $u_1$  is chosen such that the **variance of the projected data** on  $u_1$  is maximized.

The component  $u_2$  is then chosen using the projection on  $\text{Proj}_{u_1^\perp}(X)$  (orthogonal to  $u_1$ ).

### Mathematical Model:

The empirical mean and projected variance are:

$$x = \frac{1}{N} \sum_{i=1}^N x_i$$

$$v_{u_1} = \frac{1}{N} \sum_{i=1}^N (u_1^T x_i - u_1^T x)^2 = u_1^T \Sigma u_1$$

where  $\Sigma = \frac{1}{N} \sum_{i=1}^N (x_i - x)(x_i - x)^T$  is the **covariance matrix** of the data.

### Optimization Problem:

$$u_1 = \arg \max_{u^T u = 1} u^T \Sigma u$$

## Solution via Lagrange Multipliers

### Lagrangian Function:

$$J(u) = u^T \Sigma u + \lambda(1 - u^T u)$$

### Optimality Conditions:

$$\left. \frac{\partial J(u)}{\partial u} \right|_{u=u_1} = 0 \quad \text{and} \quad \left. \frac{\partial J(u)}{\partial \lambda} \right|_{\lambda=\lambda_1} = 0$$

### Fundamental Result:

$$\Sigma u_1 = \lambda_1 u_1 \quad \Rightarrow \quad v_{u_1} = u_1^T \Sigma u_1 = \lambda_1$$

**Conclusion:**  $(u_1, \lambda_1)$  is an **eigenpair** of the covariance matrix  $\Sigma$

## Complete Decomposition

Continuing this decimation process, we obtain the set of **principal components** as the eigenpairs  $\{(u_i, \lambda_i)\}_{i \in [D]}$  of  $\Sigma$ .

### Matrix Decomposition:

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_D \end{pmatrix} = \begin{pmatrix} | & | & \cdots & | \\ u_1 & u_2 & \cdots & u_D \\ | & | & \cdots & | \end{pmatrix}^T \Sigma \begin{pmatrix} | & | & \cdots & | \\ u_1 & u_2 & \cdots & u_D \\ | & | & \cdots & | \end{pmatrix} = U^T \Sigma U$$

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## Minimizing Error

### Variance-Error Equivalence

The variance  $v_{u_1}$  can be expressed as a sum of squares:

$$v_{u_1} = \frac{1}{N} \sum_{i=1}^N (u_1^T (x_i - x))^2$$

### Reasoning via the Pythagorean Theorem:

- To maximize  $v_{u_1}$ , the terms  $u_1^T (x_i - x)$  must be collectively maximized
- Since  $(x_i - x)$  is fixed, this is equivalent to **minimizing the distance to the projection axis** (approximation error)

### Alternative Formulation:

$$u_1 = \arg \min_{u^T u = 1} \sum_{i=1}^N \|(x_i - x) - [u^T (x_i - x)]u\|_2^2 \quad \Rightarrow \quad \Sigma u_1 = \lambda_1 u_1$$

### Fundamental Equivalence:

Maximizing projection variance    Minimizing approximation error

### Interpretation: Quadratic Regression

A principal component is a **quadratic regression** on the data

## Physical Interpretation

Consider a physical system  $X$  with masses  $m_i = 1$  at positions  $x_i$ .

### System Inertia:

The inertia with respect to a point  $a \in \Omega$  is:

$$I_a(X) = \sum_{i=1}^N d^2(a, x_i)$$

### Huygens' Theorem:

If  $g = \frac{1}{N} \sum_{i=1}^N x_i$  (center of gravity), then:

$$I_a(X) = d^2(a, g) + I_g(X)$$

This relation is the physical analog of  $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ .

**For a subspace  $F$  passing through  $g$ :**

$$I_F(X) = \sum_{i=1}^N d^2(F, x_i) \quad \text{where} \quad d(F, x_i) = \|x_i - \text{Proj}_F(x_i)\|$$

**Physical Conclusion:** A principal component is a **subspace of minimal inertia** with respect to  $X$

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## Dimensionality Reduction

### Structure of the Latent Space

### Properties of the Latent Space

The latent space with basis  $\{u_1, \dots, u_D\}$  has the following properties:

### Eigenvalue Order:

By construction:  $\lambda_1 > \lambda_2 > \dots > \lambda_D$

### Total Variance:

$$\lambda \stackrel{\text{def}}{=} \sum_{d=1}^D \lambda_d = \text{Tr}(\Sigma)$$

$\lambda$  represents the **total variance** of the data.

**Decorrelation of Latent Features:**

$$v_{u_d} = \lambda_d \quad \text{and} \quad u_d^T u_{d'} = 0 \text{ for } d \neq d'$$

$\Lambda$  is the **diagonal covariance matrix** in the latent space.

**Latent Coordinates**

The basis  $\{u_1, \dots, u_D\}$  induces **latent coordinates**  $y_i$  for the data:

$$y_i(d) = \langle u_d, x_i - x \rangle = u_d^T (x_i - x)$$

In vector form:

$$y_i = U^T (x_i - x)$$

**PCA Linear Transformation**

The PCA transformation is **linear** and consists of three steps:

1. **Centering** on  $x$
2. **Rotation** using matrix  $U$
3. **Scaling** (whitening) using  $\Lambda^{1/2}$



**Underlying Model**

**Variance Preservation:**

The latent space preserves the variance of the centered data.

**Distribution Assumption:**

$$X_i \sim f_X = \mathcal{N}(x, \Sigma)$$

**Important Limitation:** PCA will **not be relevant for non-Gaussian data** (e.g., clustered data)

## Exact Transformation

**Crucial Point:** So far, PCA is an **exact transformation**

$$y_i = U^T(x_i - x) \quad \text{thus} \quad Uy_i = UU^T(x_i - x) = x_i - x$$

Since  $U$  is orthogonal:  $UU^T = I_D$

At this stage, the goal is to study the **spectrum**  $\{\lambda_1 > \lambda_2 > \dots > \lambda_D\}$  of the data to understand how variance is distributed.

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## Low-Rank Approximation and Dimensionality Reduction

### Eckart-Young Theorem

**Fundamental Theorem:**

If  $\Sigma = U\Lambda U^T$  and for  $K < D$ , we define:

$$\Sigma_K \stackrel{\text{def}}{=} \sum_{d=1}^K \lambda_d u_d u_d^T$$

Then:

$$\arg \min_{\text{rank}(S)=K} \|\Sigma - S\|_F^2 = \Sigma_K$$

where  $\|\cdot\|_F$  is the **Frobenius norm**.

**Approximation Error:**

$$\|\Sigma - \Sigma_K\|_F^2 = \sum_{d=K+1}^D \lambda_d$$

$\Sigma_K$  is the **closest rank- $K$  matrix** to  $\Sigma$

## Practical Truncation

Truncation of  $U$  and  $\Lambda$ :

$$\Sigma_K = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_K \\ | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_K \end{pmatrix} \begin{pmatrix} | & & | \\ u_1 & \cdots & u_K \\ | & & | \end{pmatrix}^T = U_K \Lambda_K U_K^T$$

**Projection and Reconstruction:**

- **Projection** (encoding):  $\tilde{y}_i = U_K^T x_i \in \mathbb{R}^K$
- **Reconstruction** (decoding):  $\tilde{x}_i = U_K U_K^T x_i + x$

**Important:** We go from  $D$  dimensions to  $K$  dimensions with  $K < D$

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## Geometry and Reconstruction Quality

### Relative Contribution to Variance

Contribution of Each Dimension:

$$c_d = \frac{\lambda_d}{\sum_k \lambda_k}$$

This metric depends on the **spectrum distribution** of eigenvalues.

### Projection Ratio

Projection ratio of a data point  $x_i$  on a latent factor:

$$\rho_d(x_i) = \frac{\langle u_d, x_i \rangle^2}{\|x_i\|^2} = \cos^2(\angle(u_d, x_i))$$

**Interpretation:**

- The closer  $\rho_d(x_i)$  is to 1, the more  $x_i$  lies on the axis  $u_d$
- These metrics can be **grouped** (summed) to evaluate quality with respect to a subspace  $\{u_1, u_2, \dots\}$



## Normalization and Scaling

### Problem of Different Scales:

Each original dimension has its own unit (scale), estimated by  $\sigma_d^2$  (empirical variance along dimension  $d$ ).

PCA is more effective if **all scales are similar**

### Solution: Normalization

We create the **scaling matrix**:

$$S = \text{diag}[\sigma_1^2, \dots, \sigma_D^2]$$

We define the **Mahalanobis metric**:

$$d_S^2(x, y) = (x - y)^T S^{-1} (x - y)$$

In this metric space, the covariance matrix  $\Sigma_S$  is also normalized (it becomes the **correlation matrix**) and used as the basis for PCA.

**Practical Tip:** Normalizing data before PCA often improves results

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## Practical Application of PCA

### Complete PCA Algorithm

#### Application Steps:

Given dataset  $X \in \Omega$ :

1. **Compute** the empirical mean  $x = \frac{1}{N} \sum_{i=1}^N x_i$
2. **Center** the data:  $x_i \leftarrow (x_i - x)$  and form the centered matrix  $X$
3. **Compute** the covariance matrix:  $\Sigma = \frac{1}{N} X X^T$
4. **Decompose** into eigenvalues:  $\Sigma = U \Lambda U^T$

So far: **exact** transformation

5. **Select** the number of components  $K$
6. **Define**  $U_K$ ,  $\Lambda_K$  and compute  $\{\tilde{y}_i\}_{i \in \llbracket N \rrbracket}$  and/or  $\{\tilde{x}_i\}_{i \in \llbracket N \rrbracket}$

## Choosing the Number of Components $K$

### Three Main Strategies:

#### Strategy 1: Target Dimension

$$K = d$$

where  $d$  is the desired dimension, giving  $\tilde{y}_i \in \mathbb{R}^d$

#### Strategy 2: Variance Threshold

Require that the latent space retains a proportion  $\tau$  of the total variance:

$$\text{Var}(Y) = \tau \cdot \text{Var}(X) \quad \Rightarrow \quad K \text{ such that } \frac{\sum_{d=1}^K \lambda_d}{\sum_{d=1}^D \lambda_d} \geq \tau$$

**Example:**  $\tau = 0.95$  means retaining 95% of the variance

#### Strategy 3: Reconstruction Error

Train  $K$  such that  $L(X) \leq \epsilon$ , where for example:

$$L(X) = \sum_i \|x_i - \tilde{x}_i\|^2$$

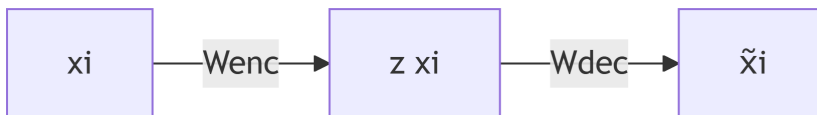
**Practical Advice:** Visualizing the eigenvalue spectrum (scree plot) helps identify a natural “elbow”

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## Alternatives and Extensions

### Linear Autoencoders (Linear AE)

#### Structure of a Linear Autoencoder:



- $W_{\text{enc}}$  and  $W_{\text{dec}}$ : weight matrices (encoder and decoder)
- $z(x_i) = W_{\text{enc}}x_i$ : encoding

- $\tilde{x}_i = W_{\text{dec}} z(x_i)$ : decoding

### Optimization Problem:

$$\theta^* = \arg \min_{W_1, W_2} \sum_{i=1}^N \frac{1}{2} \sum_{d=1}^D (x_i[d] - \tilde{x}_i[d])^2 = \arg \min_{\text{rank}(W)=K} \|X - WZ\|_F^2$$

### Relationship Between PCA and Linear Autoencoder

#### SVD Decomposition:

Using the Eckart-Young theorem with  $X = U\Psi V^T$ , the solution is:

$$W_{\text{dec}} Z = U_K \Psi_K V_K^T$$

#### Optimal Solution:

Setting  $W_{\text{dec}} = U_K \Psi_K$ , then:

$$W_{\text{enc}} = \Psi_K^{-1} U_K^T$$

Reconstruction:

$$\tilde{x}_i = U_K U_K^T x_i$$

#### Mathematical Relationship:

For **centered** data:

- **PCA:**  $\Sigma = \frac{1}{N} X X^T = U \Lambda U^T$  thus  $\tilde{x}_i = U_K U_K^T x_i$
- **AE:**  $X = U \Psi V^T$  thus  $\Sigma = \frac{1}{N} U \Psi^2 U^T$  and  $\tilde{x}_i = U_K U_K^T x_i$

Relationship between eigenvalues:

$$\Lambda = \frac{1}{N} \Psi^2 \quad \Rightarrow \quad \text{PCA}(X) \leftrightarrow \text{AE} \left( \sqrt{\frac{1}{N}} X \right)$$

**Fundamental Conclusion:** A linear autoencoder performs PCA if the data is **centered and normalized**

**Note:**  $W_{\text{enc}}$  is **not the inverse** of  $W_{\text{dec}}$  if  $K < D$

## Extensions of Autoencoders

### Nonlinear Autoencoders:

By adding hidden layers with nonlinear activation functions, the latent space becomes **nonlinear**.

### Variational Autoencoders (VAE):

By changing the loss function (using ELBO - Evidence Lower Bound), we can impose **sparsity** in the latent space.

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## Alternative Formulations of PCA

### Probabilistic PCA

#### Reformulation with Explicit Latent Distribution:

- Latent distribution:  $\mathbb{P}(z) = \mathcal{N}(0, I_{dK})$
- Conditional model:  $\mathbb{P}(x|z) = \mathcal{N}(Wz + \mu, \sigma^2 I_{dD})$

#### Advantages:

- Accommodation of **measurement noise** (variance  $\sigma^2$ )
- **Generative mode**: possibility to generate new data
- Parameter estimation by **Maximum Likelihood**
- **EM algorithm** (Expectation-Maximization) to save computations
- **Bayesian PCA**: inversion of the conditional to find  $K$  by training

### Kernel PCA

#### Principle:

Use a **nonlinear mapping**  $\phi(x_i)$  via a kernel function:

$$k(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Allows performing PCA in a **more favorable space** (possibly infinite-dimensional) without explicitly computing  $\phi$ .

## Local PCA

Performs PCA on **local neighborhoods** of the data to consider only the **local intrinsic dimensionality**.

Useful for data with complex local structure (nonlinear manifolds).

## Computational Limitation

**Complexity:** The main limitation of classical PCA is the **eigendecomposition** of  $\Sigma$  in  $O(D^3)$

For very large dimensions, alternative or approximate methods may be necessary.

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## Synthesis and Key Points

### Essential Characteristics of PCA

#### Nature of the Model:

- PCA is part of **linear latent models**
- Applies to **centered data**
- Uses **variance** as a decomposition criterion

#### Mathematical Properties:

- **Exact and complete** decomposition into **decorrelated** components
- Assumes a **normal distribution** of data
- Can be formulated as a **regression**

#### Applications:

- **Justified decimation** strategy based on low-rank approximation
- Useful for **denoising** via a Gaussian noise model

#### Equivalences:

- Equivalent to a **linear autoencoder**
- Can receive a **stochastic formulation**
- Generalizable to the **nonlinear case via kernels**

## When to Use PCA?

### Suitable for:

- Data approximately following a Gaussian distribution
- Dimensionality reduction with variance preservation
- Visualization of high-dimensional data
- Denoising data corrupted by Gaussian noise
- Data compression

### Not Suitable for:

- Strongly clustered data (non-Gaussian)
- Data with strongly nonlinear relationships
- Cases where variance is not a good information criterion

## Mathematical Concepts to Master

To understand PCA well, you need to master:

- **Linear transformation** and **orthogonal matrix**
  - **Coordinates** and **matrix rank**
  - **Mean** and **variance**
  - **Orthogonal projection**
  - **Eigendecomposition**
  - **Matrix trace**
  - **Frobenius norm**
  - **Lagrange multipliers**
- 

## Typical Exam Questions

### Conceptual Questions

1. Explain how PCA uses variance as a criterion.
2. Show that maximizing variance is equivalent to minimizing error.
3. Show how PCA uses the Eckart-Young theorem.
4. Can PCA be applied to any data?
5. Is it relevant to apply PCA to any data?

6. How to apply PCA to clustered data?

### Technical Questions

Given data:

- How do you apply PCA?
- What information does PCA provide?
- How to reconstruct data with  $K < D$  components?
- How to select the components to retain?

### Mathematical Proofs

- Show that PCA is equivalent to a linear autoencoder.
- Demonstrate the equivalence between maximum variance and minimum error.
- Prove that principal components are the eigenvectors of  $\Sigma$ .